

Category prediction Binomial logistic regression estimates the probability of an event (in this case, Host of the Show) occurring. It is very common to use binomial logistic regression to predict whether cases can be correctly classified (i.e., predicted) from the independent variables. Therefore, it becomes necessary to have a method to assess the effectiveness of the predicted classification against the actual classification. There are many methods to assess this with their usefulness often depending on the nature of the study conducted. However, all methods revolve around the observed and predicted classifications, which are presented in the "Classification Table", as shown below: Firstly, notice that the table 3 has a subscript which states, "The cut value is .500". This means that if the probability of a case being classified into the "Trevor Noah" category is greater than .500, then that particular case is classified into the "Trevor Noah" category. Otherwise, the case is classified as in the "John Oliver" category (as mentioned previously). Whilst the classification table appears to be very simple, it actually provides a lot of important information about your binomial logistic regression result, including: the model explained the John Oliver category 92.3% and Trevor Noah category 16.7% correctly predict when overall model accuracy percentage is 62.4%. Variables in the equation The "Variables in the Equation" table shows the contribution of each independent variable to the model and its statistical significance. This table is shown below: The Wald test ("Wald" column) is used to determine statistical significance for each of the independent variables. The statistical significance of the test is found in the "Sig." column. From these results you can see that Pitch Level ( $p = 0.000$ ), and Pitch Variation ( $p = 0.001$ ) added significantly to the model/prediction. From the odds ratio of pitch level is 1.009, we can say that for per unit increment of pitch level, the probability of host being Trevor Noah and John Oliver is almost similar. Table 4 also shows that the odds ratio of pitch variation is 0.229, which claim that for per unit increment of pitch variation, the probability of host being Trevor Noah is 0.771 time less likely than John Oliver. You can use the information in the "Variables in the Equation" table to predict the probability of an event occurring based on a one unit change in an independent variable when all other independent variables are kept constant. The Logistic Regression Equation:  $\text{Logit}(p) = -1.756 + 0.009 * \text{Pitch Level} - 1.475 * \text{Pitch Variation}$